Behavioral Portfolio Optimizer

A quantitative finance project that integrates traditional portfolio optimization with behavioral finance principles to construct psychologically robust investment portfolios using real market data.

Project Overview

Traditional portfolio models assume investors are fully rational and risk is captured by variance. However, real-world investors exhibit loss aversion, asymmetric risk perception, and emotional reactions during market downturns.

This project extends classical portfolio optimization by incorporating behavioral utility based on Prospect Theory to account for investor psychology in asset allocation decisions.

Objectives

- Build a traditional Mean-Variance optimized portfolio
- Develop a Behavioral Portfolio Optimizer using asymmetric loss modeling
- Compare risk-return characteristics of both approaches
- Evaluate emotional risk reduction in portfolio allocation

The traditional framework is based on
Modern Portfolio Theory

The behavioral extension is inspired by
Prospect Theory

Dataset

Historical equity data was programmatically fetched using the Yahoo Finance API for selected constituents of the
NIFTY 50.

Data used:

- 10 Large-cap NIFTY 50 stocks

- Daily historical prices (2019–2024)
- NIFTY 50 Index for benchmarking
- 252 trading days annualization assumption

No static CSV datasets were used — the pipeline dynamically retrieves live market data.

Tech Stack

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy (SLSQP optimization)
- yfinance API

Exploratory Data Analysis

EDA included:

- Price trend visualization
- Daily return distributions
- Correlation heatmap
- Volatility comparison
- Cumulative growth analysis
- Portfolio vs Index performance comparison

Key observations:

- Volatility clustering during stress periods
- High cross-sector correlation during market downturns
- Fat-tailed return distributions

These findings justify incorporating behavioral loss modeling.

Methodology

1️⃣ Data Preparation

- Fetch stock prices
- Compute daily returns
- Annualize mean returns and covariance matrix

2️⃣ Traditional Optimization

Objective:

Minimize portfolio variance subject to:

- Sum of weights = 1
- Long-only constraint ($0 \leq$ weights ≤ 1)

Efficient Frontier generated using random portfolio simulation.

Metrics:

- Expected Return
- Volatility
- Sharpe Ratio

3️⃣ Behavioral Optimization

Instead of maximizing Sharpe ratio, we maximize behavioral utility:

Prospect Theory Value Function:

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0 \\ -\lambda(-x)^{\beta} & x < 0 \end{cases}$$

Where:

- $\lambda = 2.25$ (loss aversion coefficient)
- $\alpha = \beta = 0.88$

Objective:

Maximize expected behavioral utility of portfolio returns relative to reference return.

Optimization performed using constrained nonlinear programming (SLSQP).

Results & Comparison

Metric	Traditional	Behavioral
Expected Return	Slightly Higher	Slightly Lower
Volatility	Higher	Lower
Downside Sensitivity	Not Explicit	Explicitly Penalized
Emotional Stability	Moderate	Improved

Key Insight:

The behavioral portfolio reduces exposure to high-volatility assets and limits extreme downside risk, leading to more psychologically sustainable investment outcomes.

Key Takeaways

- Risk is not purely statistical — it is psychological.
- Variance alone does not capture investor discomfort.
- Behavioral optimization produces more stable allocation under stress.
- Integrating finance theory with behavioral economics enhances portfolio realism.

Sample Visualizations

- Efficient Frontier
- Correlation Heatmap
- Volatility Bar Chart
- Cumulative Portfolio Growth
- Traditional vs Behavioral Comparison

Conclusion

This project demonstrates the integration of quantitative finance and behavioral economics to construct investor-aware portfolio strategies.

By incorporating asymmetric loss modeling into optimization, the framework moves beyond purely mathematical efficiency and aligns portfolio construction with real-world investor behavior.

The approach is applicable in:

- Robo-advisory systems
- Wealth management platforms
- FinTech risk profiling engines
- Behavioral investment research

Future Improvements

- Dynamic regime detection (Bull/Bear markets)
- Sortino-based downside optimization
- Reinforcement learning asset allocation
- Bayesian portfolio estimation
- Sentiment-driven behavioral calibration

Author

Anjali Joshi
Data Science | Quantitative Finance | Behavioral Modeling